MIL-A-48316(PA)
1 April 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

AMPLIFIER, TRIGGER PULSE: M4E1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of amplifier; designated Amplifier, Trigger Pulse: M4E1; which is used to amplify a voltage signal from the M35 impact fuze set, for warhead detonation, in the Pershing warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

FSC 1135

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DRAWINGS

PICATINNY ARSENAL

8808362 - Packing and Marking for Storage and Shipments.
8848786 - Drum, Metal, Shipping, for Amplifier, Trigger Pulse.
9287934 - Amplifier, Trigger Pulse: M4E1.
9288593 - Schematic Diagram, Amplifier: M4E1.
10625852 - Case, Top.
10625853 - Case, Bottom.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence. See 6.2) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. - The amplifier shall comply with all requirements specified on drawing (dwg.) 9287934, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article inspection. - This specification contains technical provisions for first article inspection (see 4.3). Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.4 Continuity. - Continuity shall exist between connector terminals P1A and P1B (see dwg. 9288593), and the resistance shall be a maximum of 0.05 ohm. Testing shall be as specified in 4.5.1.

3.5 Insulation resistance. - The insulation resistance of the amplifier shall be not less than 500 megohms between the connector pins listed in Table I, and the polarity of the test voltage shall be as specified in Table I. After humidity testing, the insulation resistance of the amplifier shall be not less than 20 megohms. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.2.

TABLE I

From P1 (+)	To P1 (-)
A-B	All other pins, case, and connector shell
C-F-H	"
D-E	"
G	"

3.6 Amplifier operational tests.

3.6.1 Trigger input voltage.- The amplifier shall trigger (as evidenced by a rise in the voltage output curve) when subjected to a positive peak input pulse of 3.0 ± 0.2 volt for a duration of 5.2 ± 0.2 microsecond (usec). Testing shall be as specified in 4.5.3.1.

3.6.2 Amplifier output voltage.- The amplifier's output voltage shall be a positive pulse (P1D(+), P1E(-)), and its amplitude shall be a minimum of 20 volts for a minimum duration of 15 usec after triggering. The 15-usec duration shall begin within a 5-usec period after triggering. In addition, the actual output voltage for each amplifier, as specified in Table II, shall be recorded (see 3.12). Testing shall be as specified in 4.5.3.2. If the amplifier does not comply with these requirements, 3.6.3 shall apply.

TABLE II

Amplifier Output Voltage (at 5 usec after triggering)	Test Identification Number (see 3.12)*
20 min.	062001

*Not required during the following tests: high temperature, low temperature, temperature shock, humidity, and vibration.

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3.6.3 Alternate amplifier output voltage. - The amplifier's output shall be a positive pulse (P1D(+), P1E(-)), and the output energy shall be not less than 80,000 ergs within a period of 50 usec after triggering during tests conducted at room temperature and during the high temperature test. During the low temperature test, the output energy shall be not less than 70,000 ergs within a period of 50 usec after triggering. Testing, and the method of computing output energy, shall be as specified in 4.5.3.3.

3.7 Low temperature.- The amplifier shall withstand a low temperature of -33° ± 3°F., for a minimum period of 4 hours, without evidence of cracking, distortion, or other damage that may impair its intended operation. While at the low temperature (at the end of the low-temperature exposure period), the amplifier shall comply with the requirements of 3.6. After this test (at room temperature), the amplifier shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.4.

3.8 High temperature. - The amplifier shall withstand a high temperature of 160° ± 5°F., for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. While at the high temperature (at the end of the high-temperature exposure period), the amplifier shall comply with the requirements of 3.6. After this test (at room temperature), the amplifier shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.5.

3.9 Temperature shock.- The amplifier shall withstand alternate exposures to temperatures of 160° ± 5°F., and -65° ± 5°F., for a minimum period of 4 hours at each exposure, and there shall be no evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. After this test (at room temperature), the amplifier shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.6.

3.10 Vibration. - The amplifier shall withstand vibrations as specified in Table III without evidence of deterioration or damage that may impair its intended operation. After this test, the amplifier shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.7.

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TABLE III

Frequency Range (hertz)	Displacement (D.A.) or Peak g Level (± 10% Tol.)	Minimum Time Period to Cover Freq. Range (minutes)
10-30	0.20 inch	1.75
30-1000	8 g	2.0
1000-2000	10 g	3.75

3.11 Humidity. - The amplifier shall withstand a relative humidity of 95 ± 5 percent and a temperature of 150°F. ± 5°F. without evidence of deterioration or damage that may impair its intended operation. After this test (at room temperature), the amplifier shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.8.

3.12 Serialization. - A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SARPA-QA-N-A. The letter of request shall include: (a) exact nomenclature of the item as specified on the applicable drawing, (b) contract number or procurement request order number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in the applicable paragraphs in this section.

3.13 Soldering. - All soldering shall be in accordance with MIL-STD-1460.

3.14 Workmanship. - All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. - When the contract requires first article inspection, the contractor shall submit 4 consecutively produced trigger pulse amplifiers to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed. - The first article inspection samples shall be inspected for the defects specified in 4.4.2.1, using inspection methods contained therein, and for all requirements of Section 3 (also see Table IV) using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The amplifiers shall be tested in the order specified in Table IV. If any amplifier fails to comply with the applicable requirements, the first article samples shall be rejected.

TABLE IV

	Test	Requirement Paragraph
(a)	Continuity	3.4
(b)	Insulation resistance	3.5
(c)	Amplifier operational tests	3.6
(d)	Low temperature	3.7
(e)	High temperature	3.8
(f)	Temperature shock	3.9
(g)	Vibration	3.10
(h)	Humidity	3.11

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4.4 Quality conformance inspection.

4.4.1 Inspection lot. - A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination. - A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table V). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE V

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	32	31-45	28
46-60	37	46-60	31
61-75	40	61-75	34
76-95	43	76-95	36
96-130	46	96-130	38
131-180	50	131-180	40
181-250	52	181-250	42
251-400	54	251-400	43
401 and over	56	401 and over	44

4.4.2.1 Classification of defects. - Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Case, Top (see dwg. 10625852).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting hole (2 holes)	Gage
	102. Diameter of larger mounting hole, min.	Gage

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	103. Diameter of smaller mounting hole, min	Gage
Minor:		
	201. Protective finish missing or improper	Visual
	202. Evidence of poor workmanship (see 3.14)	Visual

4.4.2.1.2 Case, Bottom (see dwg. 10625853).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting hole (2 holes)	Gage
	102. Diameter of larger mounting hole, min	Gage
	103. Diameter of smaller mounting hole, min	Gage
Minor:		
	201. Protective finish missing or improper	Visual
	202. Evidence of poor workmanship (see 3.14)	Visual

4.4.2.1.3 Amplifier, Trigger Pulse: M4E1 (see dwg. 9287934).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cable length, max	Gage
	102. Weight, max	Approved Weighing Device
	103. Ability of connector to engage and disengage with mating connector	Manual/Gage
	104. Connector pin missing or damaged	Visual
	105. Height of pins in connector (see 6.3)	Visual
	106. Case damaged (including gouges, punctures, and bent mounting flanges)	Visual
	107. Cable wire cut or damaged	Visual
	108. Soldering improper (inspect during assembly) (see 3.13)	Visual
	109. Amplifier potting improper (inspect during potting operation)	Visual

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Minor:

201. Nameplate damaged or missing.....Visual
202. Marking missing, illegible, or improper.....Visual
203. Grommet missing or damaged.....Visual
204. Cable lacing missing or improper.....Visual
205. Potting hole plate missing or improperly applied.....Visual
206. Connector potting missing or improper.....Visual
207. Protective cap on connector missing or improper.....Visual
208. Evidence of poor workmanship (see 3.14).....Visual

4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in this paragraph. The tests, on the amplifier, shall be performed in the following sequence:

Test	Test Classification
(a) Continuity	Minor
(b) Insulation resistance	Major
(c) Amplifier operational tests	Major

None of these tests is considered destructive, and samples so tested shall be returned to the lot.

4.4.3.1 Continuity (see 3.4), insulation resistance (see 3.5), and amplifier operational tests (see 3.6).- Each amplifier shall be subjected to these tests (see applicable paragraph in 4.5). If any amplifier fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests. - Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration instructions prior to use on the contract (see 6.4).

4.4.4.2 Minor defects and tests. - The acceptance inspection equipment, operating instructions, and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.5 Test methods. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F.). NOTE: During environmental tests, the amplifier's connector shall be protected by its protective cap.

4.5.1 Continuity.- The continuity of the amplifier's continuity loop (measured between connector pins (P1A and P1B) shall be measured with a Government-approved tester and shall comply with the requirements of 3.4.

4.5.2 Insulation resistance. - The insulation resistance of the amplifier shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current. The test voltage shall be applied instantaneously to the amplifier, and the insulation resistance shall be measured after a 10-second application of this test voltage. The amplifier shall comply with the requirements of 3.5 during this test.

4.5.3 Amplifier operational tests.- Prior to testing, a constant load shall be connected across the amplifier's output terminals (P1D to P1E), and it shall consist of a 1.7 +0, -0.15 ohm load resistance in series with an inductance of 7 ± 1 microhenry. The total resistance in the legs of the output load circuit (excluding the 1.7 ohm load resistance) shall be 0.40 +0, -0.05 ohm. During these tests, the trigger input and the amplifier output voltages shall be monitored by means of Government-approved oscilloscopes or other recording devices. Voltage polarities are indicated by + and - signs, as applicable.

4.5.3.1 Trigger input voltage.- A suitable direct-current source shall be used to supply 26 +0.3, -0 volt between terminals P1H(+) and P1F(-). The amplifier's input pulse source shall have an impedance of 50 ohms maximum and shall be capable of producing an output voltage pulse of 3.0 volts minimum within a period of 1 microsecond (rise time) when connected to a pure resistive load (not the amplifier). The input pulse, specified in 3.6.1, shall be applied between terminals P1C(+) and P1F(-). The amplifier shall comply with the requirements of 3.6.1 during this test.

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4.5.3.2 Amplifier output voltage. - With the amplifier connected as specified in the trigger input voltage test (see 4.5.3.1), the amplifier output voltage shall be monitored across the 1.7 ohm load resistor and shall comply with the requirements of 3.6.2.

4.5.3.3 Alternate amplifier output voltage.- The amplifier shall be triggered as specified in the trigger input voltage test (see 3.6.1), and a photograph or recording of the output voltage curve shall be obtained and analyzed to determine the acceptability of the amplifier. To satisfy the 80,000 erg requirement, the coordinates of the output voltage curve shall equal or exceed all of the coordinates of one of the three conditions specified in Table VI. To satisfy the 70,000 erg requirement, the coordinates of the output voltage curve shall equal or exceed all of the coordinates of one of the three conditions specified in Table VII. The amplifier shall comply with the requirements of 3.6.3 during this test.

TABLE VI (80,000 ergs)

Time (usec)	Condition 1	Condition 2	Condition 3
5	14.5 volts	20.0 volts	22.0 volts
10	19.0	21.5	21.0
20	20.0	18.5	18.0
30	17.5	16.0	15.5
40	15.5	13.5	12.5
50	13.5	11.5	10.5

TABLE VII (70,000 ergs)

Time (usec)	Condition 1	Condition 2	Condition 3
5	14.5 volts	19.0 volts	21.0 volts
10	18.5	20.5	21.0
20	18.5	17.5	17.0
30	16.0	14.5	14.0
40	13.5	12.0	11.5
50	11.5	10.0	9.0

4.5.4 Low temperature. - The amplifier shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The amplifier shall remain at this temperature for a minimum period of 4 hours and then tested as specified in 3.6 at the end of the low-temperature exposure period. The amplifier shall then be removed from the test

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chamber, allowed to stabilize at room temperature, and then tested for the applicable requirements of 3.4, 3.5, and 3.6. The amplifier shall comply with the requirements of 3.7 during this test.

4.5.5 High temperature. - The amplifier shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The amplifier shall remain at this temperature for a minimum period of 4 hours and then tested as specified in 3.6 at the end of the high-temperature exposure period. The amplifier shall then be removed from the test chamber, allowed to stabilize at room temperature, and then tested for the applicable requirements of 3.4, 3.5, and 3.6. The amplifier shall comply with the requirements of 3.8 during this test.

4.5.6 Temperature shock. - The amplifier shall be placed into a test chamber having an internal temperature of 160°F., and it shall remain at this temperature for a minimum period of 4 hours. At the end of this test exposure, and within a 5-minute period, the amplifier shall be transferred to a test chamber having an internal temperature of -65°F., and it shall remain at this temperature for a minimum period of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the amplifier shall be subjected to three such cycles. At the completion of the third cycle (at the end of the low-temperature exposure period) the amplifier shall be removed from the test chamber and allowed to stabilize at room temperature. The amplifier shall then comply with the requirements of 3.9.

NOTE: The test may, for convenience purposes, be interrupted at the end of a low-temperature exposure period. This is accomplished by returning the amplifier to room temperature after an exposure at high temperature of 1 hour minimum. To continue the test, start with the high-temperature exposure of the test cycle.

4.5.7 Vibration. - The amplifier shall be securely mounted to a vibration machine (the cable and connector shall be attached to the vibration machine by any convenient method and may be cushioned to prevent damage). The amplifier shall be subjected to the vibrations specified in Table III along each of three mutually perpendicular axes. The amplifier shall comply with the requirements of 3.10 during this test.

4.5.8 Humidity. - The amplifier shall be placed into a dry oven and conditioned at a temperature of 130° ± 5°F. for a

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minimum period of 24 hours. At the end of this conditioning period, the amplifier shall be transferred to a humidity chamber having a temperature of 150°F. and a relative humidity of 95 percent. The amplifier shall remain in this chamber for a minimum period of 96 hours. Following this exposure period, the amplifier shall be removed from the humidity chamber and allowed to stabilize at room temperature for a minimum period of 4 hours. The fuze shall comply with the requirements of 3.11 during this test.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation, packaging, and packing.- Preservation, packaging, and packing shall be level A or C as specified (see 6.1).

5.1.1.1 Level A and level C. - The amplifier shall be preserved, packaged, and packed in accordance with instructions contained in MIL-STD-1169.

5.1.2 Marking.- Marking shall be in accordance with MIL-STD-1169.

5.2 Spare parts.

5.2.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A.- The amplifier shall be preserved and packaged in accordance with dwg. 8848786.

5.2.1.2 Level C.- The amplifier shall be preserved and packaged in accordance with dwg. 8848786 except the barrier material may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and level C. - The amplifier shall be packed in accordance with dwg. 8808362.

5.2.3 Marking. - Marking shall be as specified in accordance with dwg. 8808362 and 8848786.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

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6.1.1 Procurement requirements:
(a) Title, number, and date of this specification.
(b) Production item or spare part procurement.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.1.2 Contract data requirements. - Data specified in 3.6.2, and contractor engineering documentation specified in 4.4.4.1, will be delivered as identified on a numbered DD form 1664 when specified on a DD Form 1423 incorporated into the contract.

6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.3 Inspection of connectors.

6.3.1 Height of pins.- In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.040 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

6.4 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SARPA-QA-T-E.

6.5 Supersession data. - This specification supersedes the requirements of Purchase Description ABMA-PD-F-976 (Rev. 2), dated 31 March 1965, with Amendment 1 dated 29 June 1973.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project No.: 1135-A140

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